



Indian Institute of Technology Bhubaneswar

School of Infrastructure

Session: Spring 2026

Finite Element Methods (CE6L314)

Date: March 31, 2026

Assignment No. 2

Total Marks: 100

Instructions:

- (1) If two or more answer scripts appear identical, each of them will be awarded ZERO.
- (2) Provide neatly drawn figures to explain the concepts behind the problems whenever possible.
- (3) Assume reasonable numerical values of the parameters for the comparison between two methods.
- (4) Provide the codes used for solving the problems and generating the plots.
- (5) Submit your answer script by April 13, 2026.

1. (a) Derive the shape functions of a 3-node triangular (T3) element using area (barycentric) coordinates.
(b) Prove that the shape functions satisfy: Partition of unity and the Kronecker delta property.
(c) Write a Julia function to compute shape functions of the T3 element at any (ξ, η) .
(d) For the visualization of T3 Shape Functions, generate a grid satisfying:

$$0 \leq \xi, \eta, \quad \xi + \eta \leq 1$$

Plot (i) 3D surface plots and (ii) Contour plots.

- (e) Comment on variation of shape functions.

[7]

2. (a) Derive the shape functions of a 6-node triangular (T6) element using area (barycentric) coordinates.
(b) Prove that the shape functions satisfy: Partition of unity and the Kronecker delta property.
(c) Write a Julia function to compute shape functions of the T6 element at any (ξ, η) .
(d) Plot all six shape functions and compare with T3 shape functions.
(e) Explain the role of mid-side nodes.

[7]

3. (a) Derive the shape functions of a 10-node triangular (T10) element using area (barycentric) coordinates.
(b) Prove that the shape functions satisfy: Partition of unity and the Kronecker delta property.
(c) Write a Julia function to compute shape functions of the T10 element at any (ξ, η) .
(d) Plot selected shape functions (corner, edge, and internal nodes).

(e) Compare T3, T6, and T10 elements in terms of smoothness and discuss computational cost vs accuracy. [7]

4. (a) Derive the shape functions of a bilinear quadrilateral (Q4) element in (ξ, η) coordinates.
(b) Prove that the shape functions satisfy: Partition of unity and the Kronecker delta property.
(c) Write a Julia function to compute shape functions of the Q4 element at any (ξ, η) .
(d) For the visualization of Q4 shape functions, use domain:

$$-1 \leq \xi, \eta \leq 1$$

Plot (i) 3D surface plots and (ii) Contour plots.

(e) Comment on variation of shape functions. [7]

5. (a) Derive the shape functions of a 9-node quadrilateral (Q9) element in (ξ, η) coordinates.
(b) Prove that the shape functions satisfy: Partition of unity and the Kronecker delta property.
(c) Write a Julia function to compute shape functions of the Q9 element at any (ξ, η) .
(d) Plot selected shape functions (corner, edge, and internal nodes).
(e) Compare Q4 and Q9 elements in terms of smoothness and discuss computational cost vs accuracy. [7]

6. (a) Derive the shape functions of a 8-node quadrilateral (Q8) serendipity element in (ξ, η) coordinates.
(b) Prove that the shape functions satisfy: Partition of unity and the Kronecker delta property.
(c) Write a Julia function to compute shape functions of the Q8 element at any (ξ, η) .
(d) Plot selected shape functions (corner and edge nodes).
(e) Compare Q8 and Q9 elements in terms of smoothness and discuss computational cost vs accuracy. [7]

7. Approximate:

$$u(x, y) = x^2 + y^2$$

(a) Plot interpolated fields using Q4, Q8 and Q9 elements and compute errors for each case.

(b) Discuss completeness and convergence. [7]

8. Develop a unified Julia function for generalized shape function code for T3, T6, T10, Q4, Q8, Q9 elements.

Use element type as input and validate the implementation. [7]

9. (a) Evaluate the integral:

$$f(x, y) = x^2 + y^2 + xy$$

over the domain $[-1, 1] \times [-1, 1]$.

(b) Compare numerical results using different Gauss-point integration schemes.

(c) Compute the error with respect to the exact solution.

(d) Discuss accuracy and convergence. [7]

10. (a) Evaluate:

$$f(x, y) = x^2 + y^2 + xy$$

over a triangular domain.

- (b) Compare numerical results using different Hammer-point integration schemes.
 (c) Compute the error with respect to the exact solution.
 (d) Discuss accuracy and convergence.

[7]

11. (a) Consider heat conduction in a rectangular plate described by the governing partial differential equation given by:

$$-\nabla \cdot (k \nabla u) = 0, \quad k = 1$$

in the domain:

$$(x, y) \in (0, 2) \times (0, 1),$$

with the essential boundary conditions: $u(0, y) = 100$, $u(2, y) = 0$. The insulated top and bottom of the plate are described by the Neumann boundary condition given by:

$$\frac{\partial u}{\partial n} = 0$$

- (a) Solve for the temperature field using Q4 and T3 elements .
 (b) Plot temperature distribution and compare the results obtained using Q4 and T3 elements.

[10]

12. Consider heat conduction in a rectangular plate described by the governing partial differential equation given by:

$$-\nabla \cdot (k(x, y) \nabla u) = f$$

where:

$$k(x, y) = 1 + x + y, \quad f = 1$$

Domain: $(x, y) \in (0, 1] \times (0, 1)$

Boundary conditions: $u = 0$.

- (a) Solve for the temperature field using Q4, Q8 and Q9 elements. Plot temperature distribution and comment on h-convergence and p-convergence of the numerical results.
 (b) Solve for the temperature field using T3, T6 and T10 elements . Plot temperature distribution and comment on the h-convergence and p-convergence of the numerical results.

[20]